Ivan Yakovlev

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Experience

Research Engineer: Speech Processing Sep. 2026 – Present
Gradium *Barcelona, Spain (Remote)*

- Developing a generic data pipeline for TTS and ASR training data (pretraining and SFT), including enriched data for TTS SFT. Processing multi-million-hour multilingual speech corpus.
- Running pretraining and SFT experiments with related data and training ablations for TTS and ASR models
- Developing the simultaneous speech-to-text translation (S2TT) system

Senior Research Engineer: Speech Processing Mar. 2025 – Sep. 2026
Palabra AI Ltd. *Barcelona, Spain*

- Solo-built production zero-shot multilingual streaming TTS system (8+ languages, voice cloning, ∼100 ms time-to-first-audio), serving as the core product for Palabra's voice AI platform
- Designed and implemented cost-efficient data pipeline producing clean pseudo-labeled speech at $0.10/hour (10–50× below industry standard), collecting 200K+ hours across 20 languages
- Architected end-to-end speech-to-speech translation system for the voice AI agent market
- Researched and developed streaming LLM-ASR model (∼640 ms latency) surpassing conformer-transducer based models in most domains
- Invented ReDimNet2 architecture — second generation of ReDimNet with improved scalability via time-pooled dimension reshaping (B0–B6 model family, 1.1M–12.3M params); oral at Interspeech 2026

Sr. Machine Learning Engineer: Speech Processing Sep. 2019 – Feb. 2025
ID R&D Inc. *Barcelona, Spain (Remote)*

- Developed an automated data pipeline generating a 200k-identity dataset, doubling ASV performance across multiple domains
- Invented new SOTA neural network architecture, called Reshape Dimensions Network (ReDimNet), 60% faster, 35–50% smaller and 15–25% more accurate than existing architectures (ResNets, CAM++, ECAPA-TDNN)
- Designed a subnetwork approach for ResNet and similar architectures, improving downstream task performance by over 50% in target metrics
- Achieved top placements in the VoxCeleb Speaker Recognition Challenges (2022, 2023) and SASV Challenge, contributing to three publications
- Optimized speaker verification training pipelines with PyTorch and TPU, achieving a 4× training speedup within the same budget

Software Engineer Feb. 2017 – Mar. 2019
EPAM Systems *Saint Petersburg, Russia*

- Developed AR/VR applications integrating deep learning vision models using Unity3D

Publications

- **Ivan Yakovlev**, Anton Okhotnikov, "ReDimNet2: Scaling Speaker Verification via Time-Pooled Dimension Reshaping", Interspeech 2026 (oral)
- **Ivan Yakovlev**, Rostislav Makarov, Andrei Balykin, Pavel Malov, Anton Okhotnikov, Nikita Torgashov, "Reshape Dimensions Network for Speaker Recognition", Interspeech 2024
- **Ivan Yakovlev**, Anton Okhotnikov, Nikita Torgashov, Rostislav Makarov, Yuri Voevodin, Konstantin Simonchik, "VoxTube: a multilingual speaker recognition dataset", Interspeech 2023
- Nikita Torgashov, Rostislav Makarov, **Ivan Yakovlev**, Pavel Malov, Andrei Balykin, Anton Okhotnikov, "The ID R&D VoxCeleb Speaker Recognition Challenge 2023 System Description", VoxSRC-23 at Interspeech 2023
- Alexander Alenin, Nikita Torgashov, Anton Okhotnikov, Rostislav Makarov, **Ivan Yakovlev**, "A subnetwork approach for spoofing aware speaker verification", Interspeech 2022
- Rostislav Makarov, Nikita Torgashov, Alexander Alenin, **Ivan Yakovlev**, Anton Okhotnikov, "ID R&D System Description to VoxCeleb Speaker Recognition Challenge 2022", VoxSRC-22 at Interspeech 2022
- **Ivan Yakovlev**, Mikhail Melnikov, Nikita Bukhal, Rostislav Makarov, Alexander Alenin, Nikita Torgashov, Anton Okhotnikov, "LRPD: Large Replay Parallel Dataset", ICASSP 2022

Awards & Competitions

- **VoxCeleb Speaker Recognition Challenge**: 1st place, open track (2022, 2023; oral at Interspeech 2023); 3rd place, closed track (2020)
- **Spoofing-Aware Speaker Verification Challenge 2022**: 1st place
- **Kaggle Competitions Master**: gold in Deepfake Detection Challenge 2019 (8/2265); silver in BirdCLEF 2023 (27/1189) and Cornell Birdcall Identification 2020 (57/1390)

Education

Saint Petersburg State University St. Petersburg, Russia
Bachelor of Science in Quantum Mechanics *2012 – 2016*

Skills

Expertise: TTS, neural audio codecs and vocoders, streaming ASR, speech translation, speaker recognition and anti-spoofing, large-scale speech data pipelines
Stack: PyTorch, vLLM, LoRA/GRPO post-training, multi-node H100/A100 training (Slurm, SkyPilot), GCP, Nebius
Languages: English (B2+), Russian (native)